TECHNOVA LTD

INFRASTRUCTURE & NETWORK SERVICES

Network implementation report

Departmental connectivity, service delivery and access control

IT	SALES	HR
192.168.10.0/24	192.168.20.0/24	192.168.30.0/24

PREPARED BY

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01 / NETWORK ARCHITECTURE

Three departments. One network.

A routed topology connecting IT, Sales and HR.

The project connects three departmental networks through a central router. It provides automatic PC addressing and access to IT server services while restricting ICMP traffic from Sales to IT.

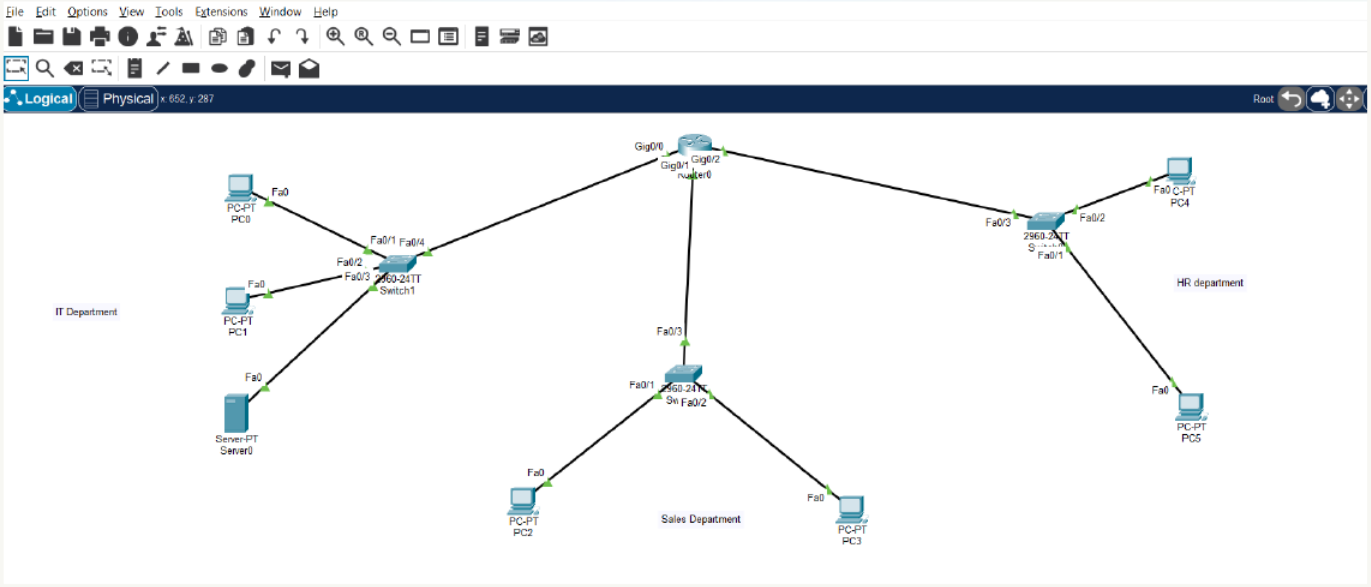


FIGURE 01 / Complete topology: one router, three switches, six PCs and one IT server.

ROUTER	DEPARTMENTAL SWITCHES	END DEVICES
01	03	07

Department	Devices shown	Subnet
IT	PC0, PC1 and Server0	192.168.10.0/24
Sales	PC2 and PC3	192.168.20.0/24
HR	PC4 and PC5	192.168.30.0/24

02 / CLIENT ADDRESSING

IT address configuration

PC1 | DHCP addressing on the IT departmental subnet.

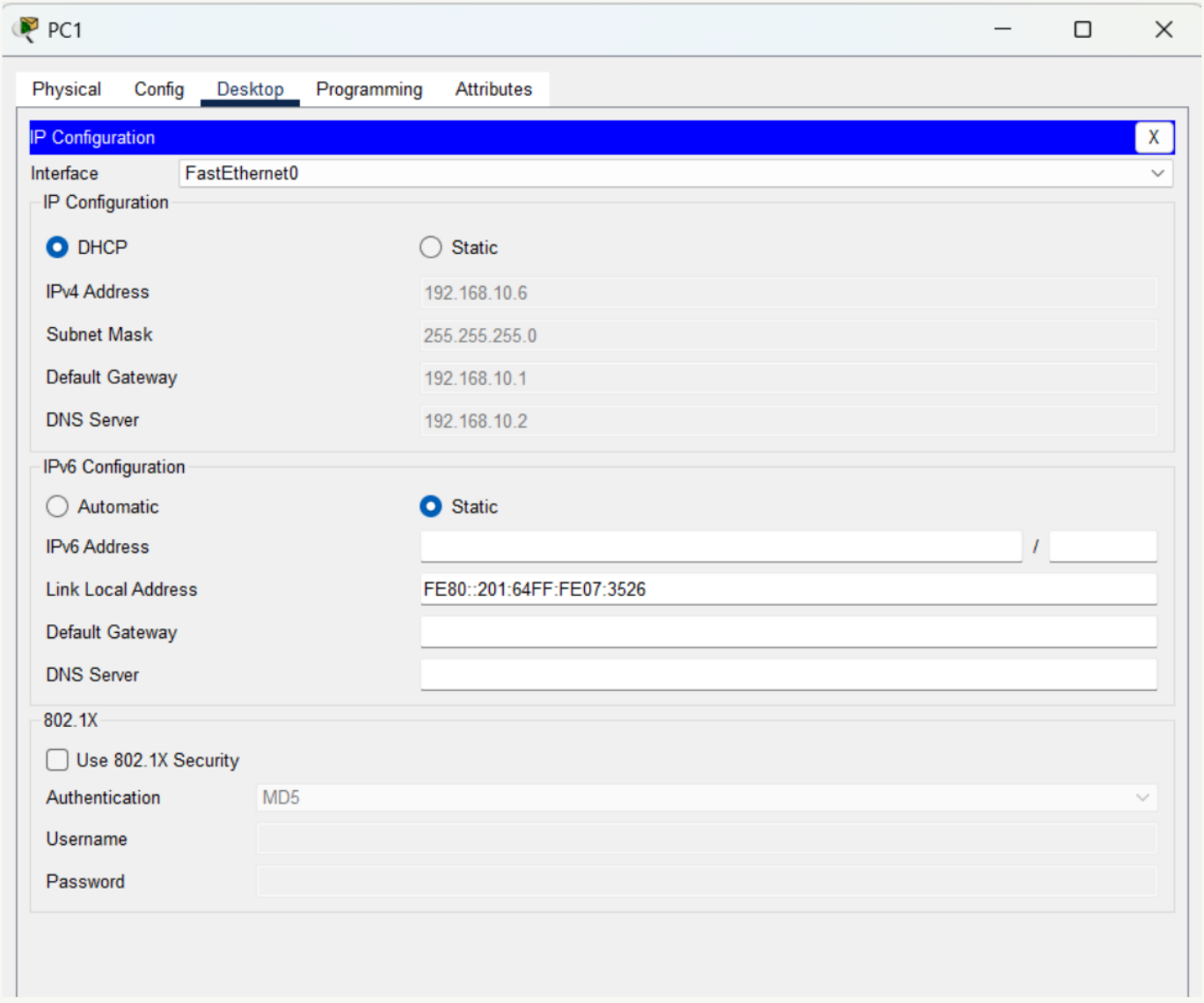


FIGURE 02 / PC1: DHCP selected, with populated IPv4 settings.

IP ADDRESS
192.168.10.6

DEFAULT GATEWAY
192.168.10.1

SUBNET MASK 255.255.255.0

DNS SERVER 192.168.10.2

02 / CLIENT ADDRESSING

Sales address configuration

PC2 | DHCP addressing on the Sales departmental subnet.

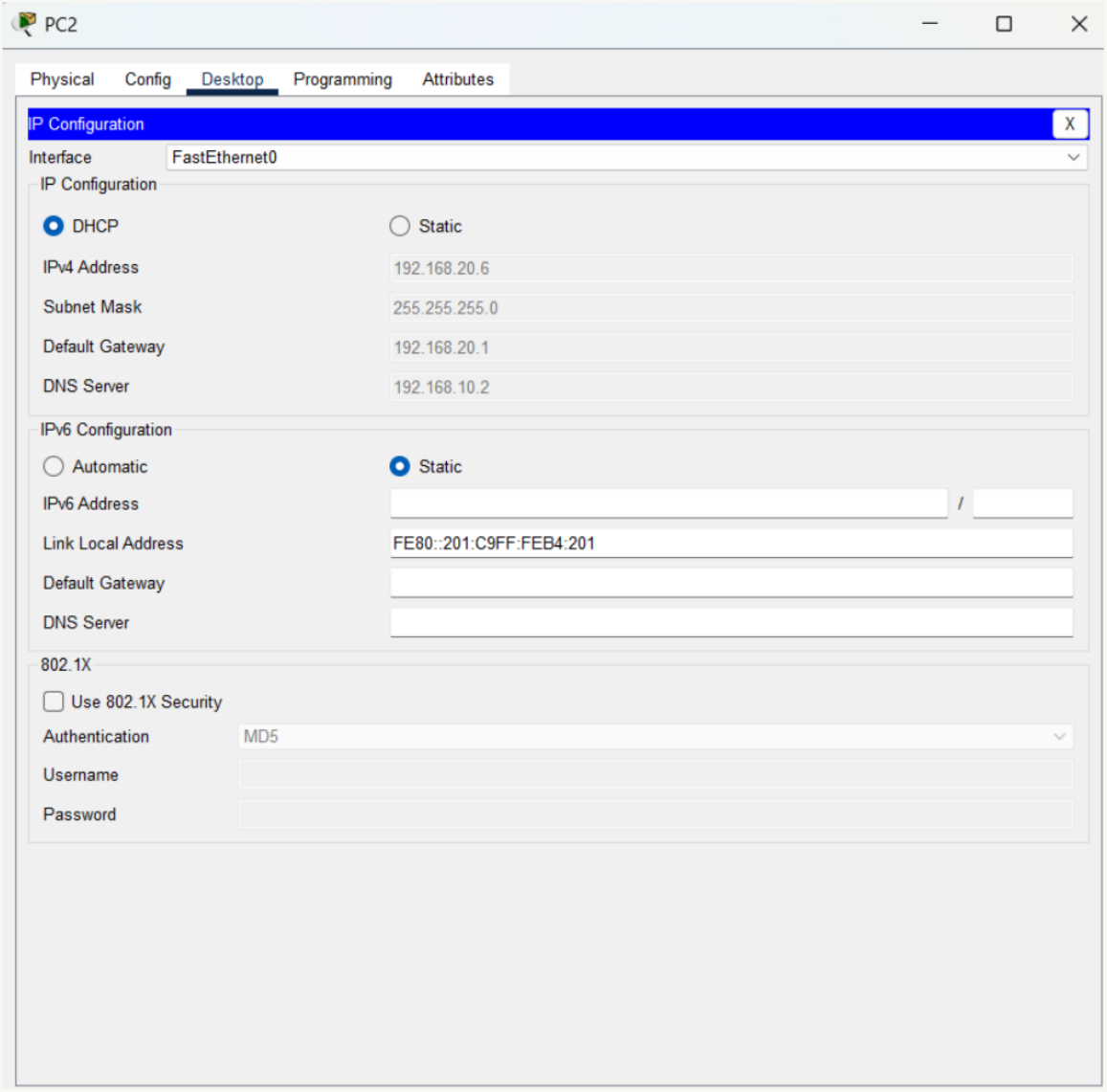


FIGURE 03 / PC2: DHCP selected, with populated IPv4 settings.

IP ADDRESS

192.168.20.6

DEFAULT GATEWAY

192.168.20.1

SUBNET MASK 255.255.255.0

DNS SERVER 192.168.10.2

02 / CLIENT ADDRESSING

HR address configuration

PC5 | DHCP addressing on the HR departmental subnet.

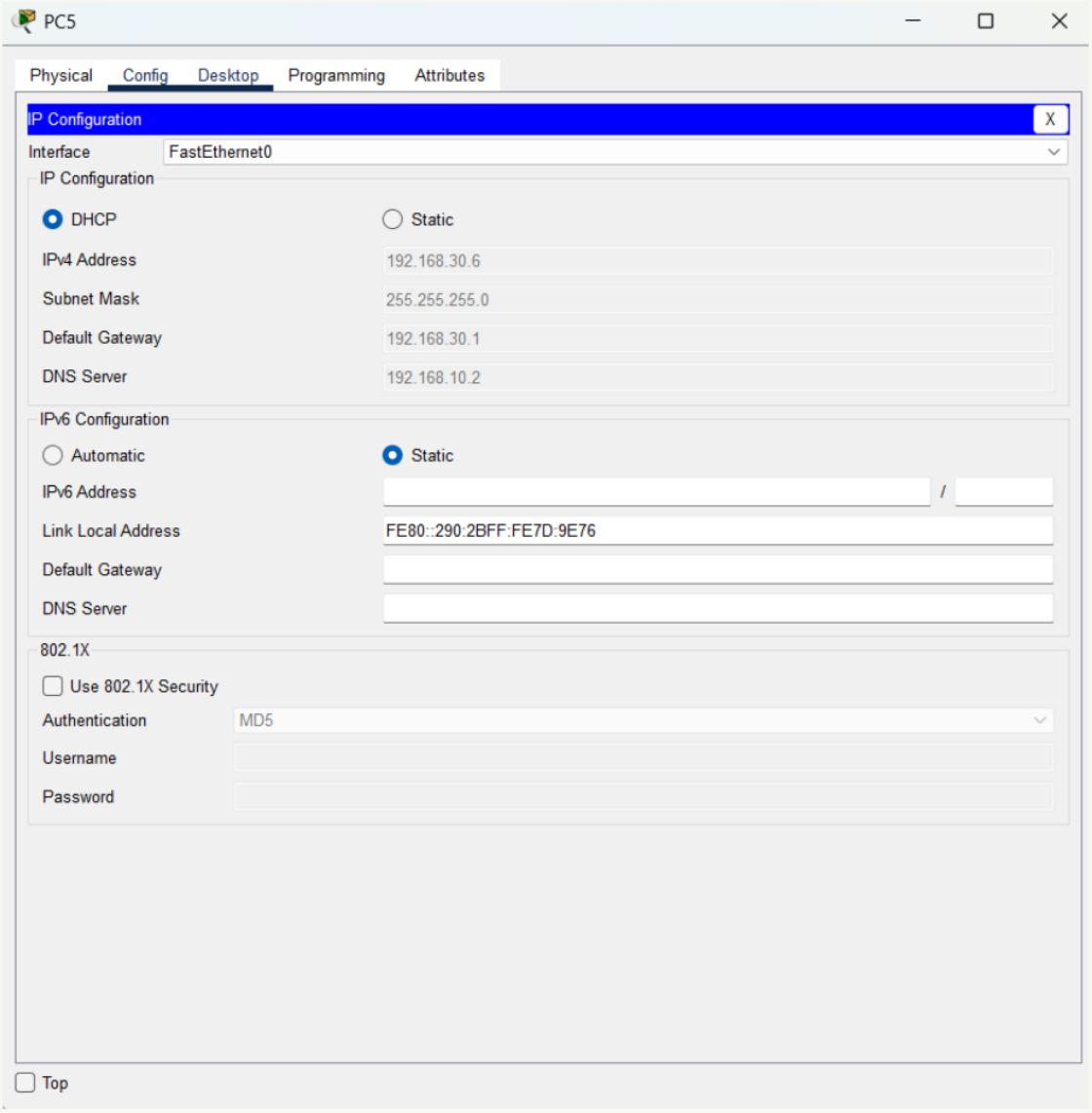


FIGURE 04 / PC5: DHCP selected, with populated IPv4 settings.

IP ADDRESS

192.168.30.6

DEFAULT GATEWAY

192.168.30.1

SUBNET MASK 255.255.255.0

DNS SERVER 192.168.10.2

03 / CENTRAL SERVICES

Addressing & name resolution

Server0 provides departmental DHCP pools and a DNS record for the website.

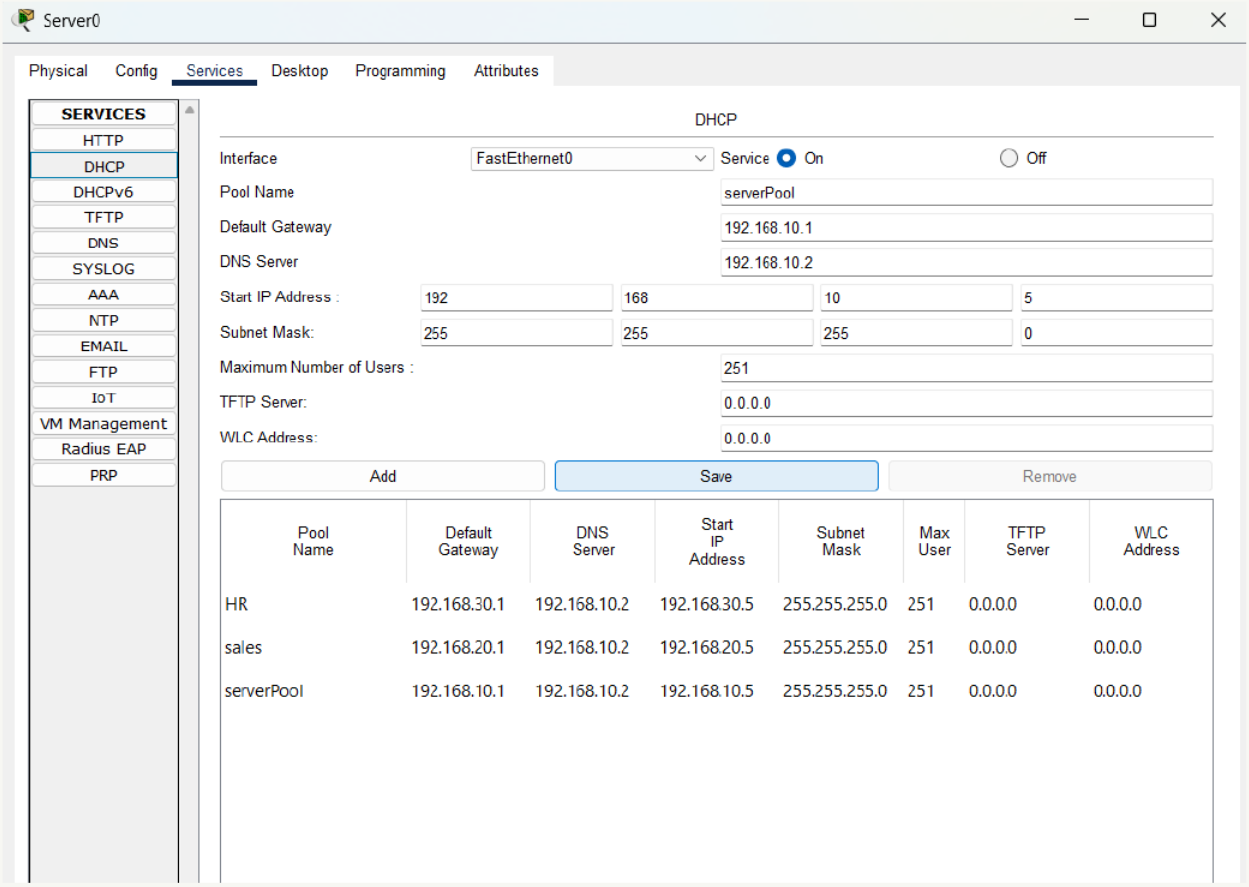


FIGURE 05 / Server0 DHCP pools: HR, sales and serverPool (IT).

All pools begin at host .5 and advertise the departmental gateway and DNS server 192.168.10.2. The configured maximum of 251 users does not limit leases to the assignment range .5-.10.

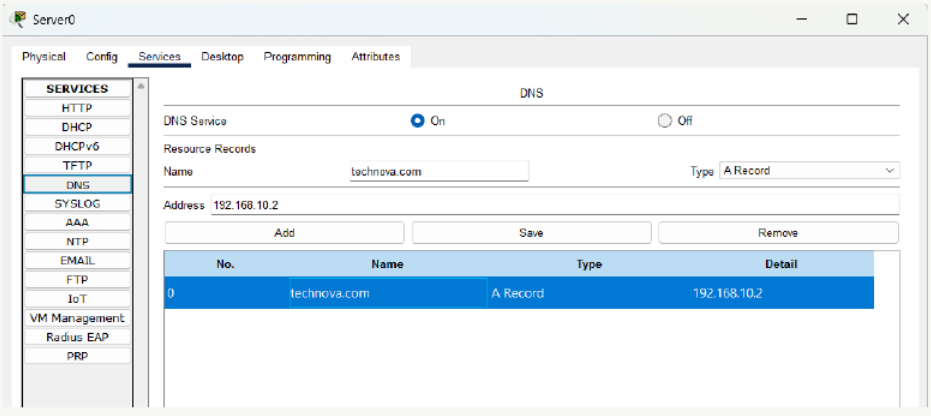


FIGURE 06 / DNS record: technova.com resolves to 192.168.10.2.

04 / ACCESS CONTROL

A targeted protocol restriction

Sales-ACL is applied inbound on the Sales-facing GigabitEthernet0/1 interface.

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip access-list extended Sales-ACL
Router(config-ext-nacl)#deny icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#int g0/1
Router(config-if)#ip access-group Sales-ACL in
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show access-lists
Extended IP access list Sales-ACL
    10 deny icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
    20 permit ip any any (5 match(es))

Router#
```

FIGURE 07 / Router CLI: configured rules, interface assignment and ACL display.

CONFIGURATION

```
ip access-list extended Sales-ACL
deny icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
permit ip any any
exit
interface g0/1
ip access-group Sales-ACL in
```

Rule behaviour

The wildcard mask 0.0.0.255 matches each /24 network. The deny rule is evaluated before permit ip any any, which allows traffic that does not match the restriction.

Actual scope

The rule blocks all ICMP from Sales to IT, including echo replies. HTTP and FTP use TCP. The displayed permit counter shows five matches; no deny count is shown.

Policy: deny Sales-to-IT ICMP; permit other IP traffic.

05 / CONNECTIVITY VALIDATION

Access control in operation

Both tests target the IT host at 192.168.10.5.

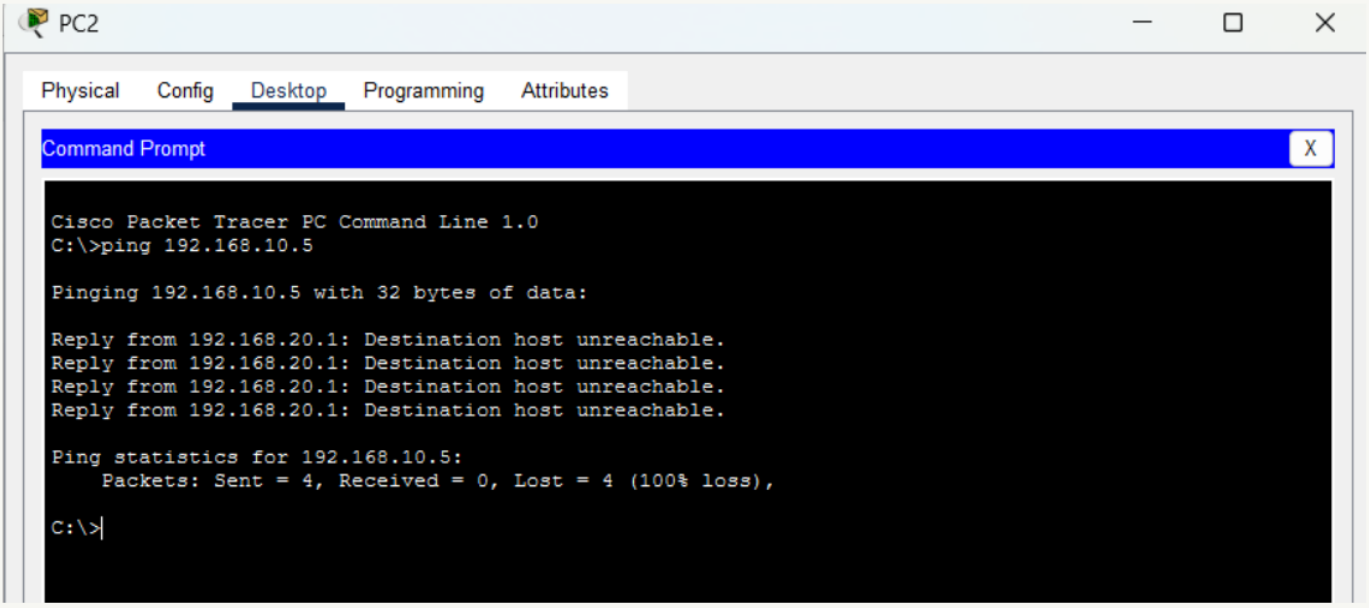


FIGURE 08 / Sales PC2: four sent, zero received, 100% loss.

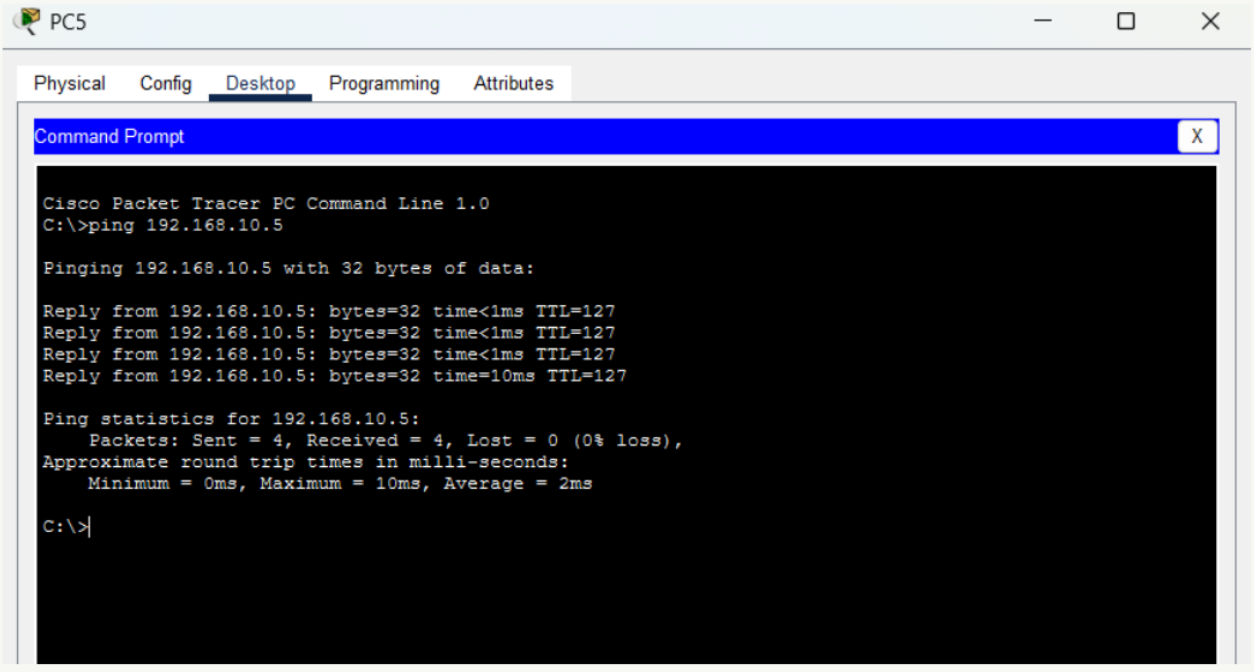


FIGURE 09 / HR PC5: four replies received, 0% loss.

These results support the Sales restriction and continued HR access. Pre-ACL and HR-to-Sales test results are not included in the screenshots.

06 / APPLICATION SERVICES

Server access remains available

TCP-based services are permitted by the configured ACL.

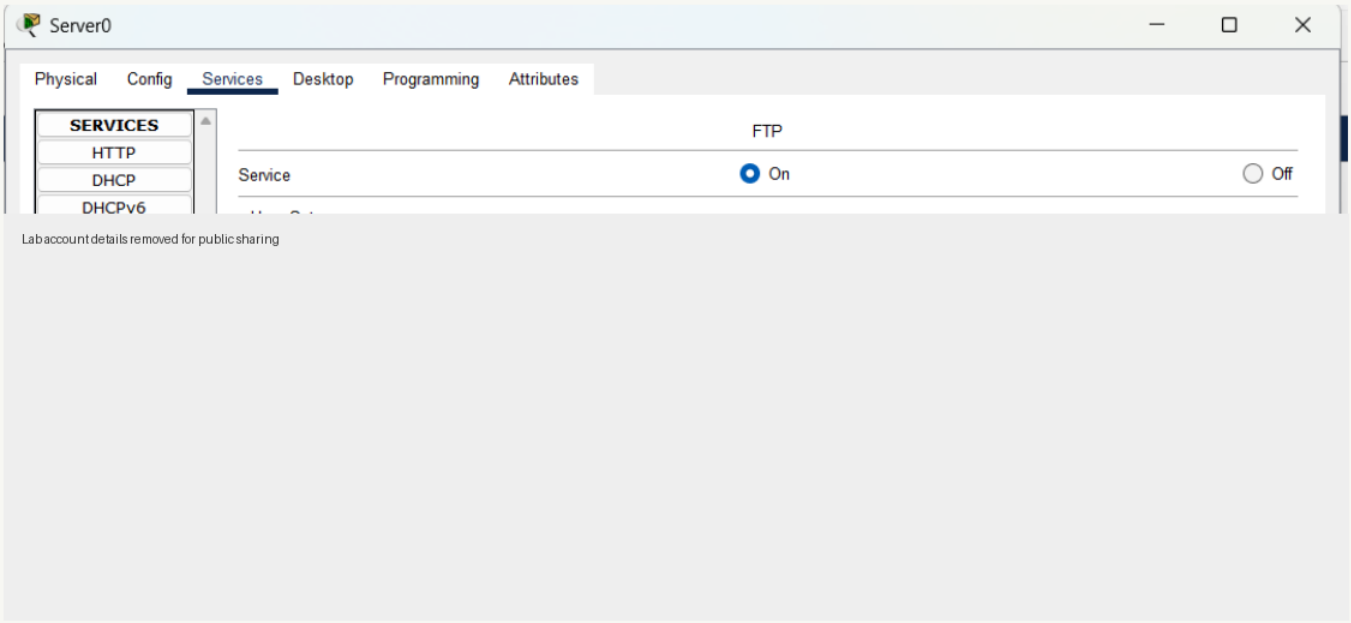


FIGURE 10 / FTP is enabled with lab accounts. A client transfer is not shown.

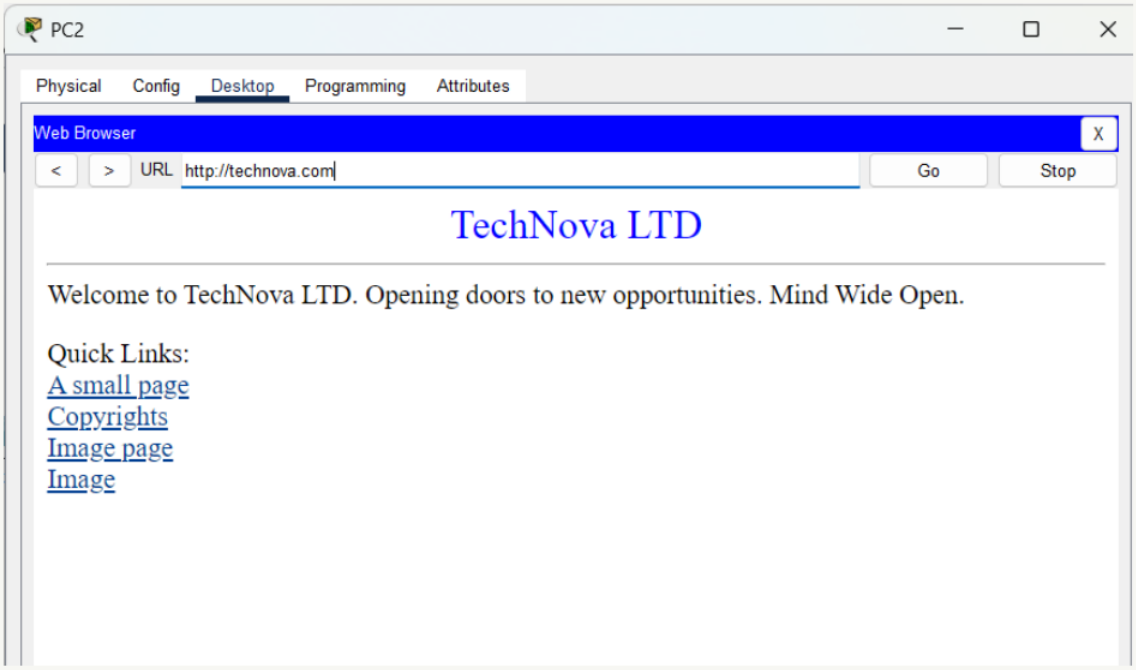


FIGURE 11 / Sales PC2 opens http://technova.com successfully.

The browser result demonstrates HTTP access from Sales. FTP is configured and permitted; no completed FTP session is shown.

07 / DESIGN RATIONALE

Why the network is structured this way

Responses to the assignment analytical questions, supported by the screenshots.

01 Why divide the network into IT, Sales and HR subnets?

Separate subnets organize departmental addressing and contain local broadcast traffic. They also create a point at the router where access between departments can be controlled. Figures 1-4 show three departments using distinct /24 networks. Separation alone does not deny access; the ACL supplies the restriction.

02 Why is a router required between the departments?

The PCs belong to different IP networks. Their local switches forward traffic within each departmental LAN, while the router forwards traffic between those networks. The PCs use the .1 address in their own subnet as the default gateway. Figure 9 demonstrates that HR can reach an IT host through the routed network.

03 Why assign a static IP address to the IT server?

A fixed server address keeps services at a predictable location. The design uses 192.168.10.2 for the server: DHCP advertises it as DNS, and the DNS A record points technova.com to it. A changing address could break these references. The screenshots support use of this address, but do not include the server's Static IP Configuration screen.

04 What is the purpose of DHCP?

DHCP automatically supplies PCs with an IP address, subnet mask, gateway and DNS settings. It reduces repetitive manual configuration and helps prevent addressing mistakes. Figures 2-4 show DHCP selected and populated settings in each department. Figure 5 shows the three pools on Server0.

08 / POLICY RATIONALE

How services remain accessible

Protocol-specific filtering explains the observed difference between ping and HTTP.

05 Why can FTP work even when ping is blocked?

Ping uses ICMP, while FTP uses TCP. The configured deny rule matches ICMP between Sales and IT; FTP traffic instead reaches the permit ip any any rule. Therefore this ACL allows FTP, provided the service, credentials and connectivity are correct. Figure 10 shows FTP enabled, but a successful FTP session is not shown.

06 What specific traffic does the ACL block?

It blocks all ICMP packets with source addresses in 192.168.20.0/24 and destination addresses in 192.168.10.0/24 when entering GigabitEthernet0/1. The screenshot does not include an echo-only qualifier. Sales-to-IT ping requests are therefore blocked, along with other matching ICMP messages.

07 Why can a Sales PC access IT HTTP/FTP services but not ping IT?

Access is decided by protocol as well as source and destination. ICMP matches the deny rule, but TCP-based HTTP and FTP are permitted by the next rule. Figure 8 shows the failed Sales ping, and Figure 11 shows the same Sales PC displaying the website. HTTP access is demonstrated; FTP access is permitted by the configuration but not independently demonstrated.

Conclusion

Departmental addressing, blocked Sales-to-IT ping, successful HR-to-IT ping and Sales HTTP access are evidenced. Align DHCP pool limits with .5-.10 if required. Further validation can include an FTP login and the remaining connectivity tests.